Submission Summary

Conference Name
6th OPJU International Technology Conference (OTCON 6.0) on SMARTICON (Smart & Intelligent Computing for Next-Generation Technologies)

Track Name
Smart Computing for IoT, Edge Computing & Intelligent Sensor Systems

Paper ID
331

Paper Title
CHROMA: Colour Intelligence-Bridging Physical Colours and Digital Design

Abstract
Transferring colours from physical surfaces into digital design applications is complicated by illumination, surface reflectance, sensor response and fragmented software workflows. This paper presents CHROMA, a prototype integrating CHROMAPOD, a portable ESP32–TCS34725 colour scanner, with CHROMA Studio, a mobile-compatible colour analysis application. CHROMAPOD supports acquisition, local CIE76 comparison and Bluetooth Low Energy (BLE) exchange; the companion application provides format conversion, CIEDE2000 image matching, surface recolouring and colour-reference management. Calibration was evaluated on 16 printed RGB reference patches, with five complete scans per patch for each of two firmware versions (80 scans per version). Mean channel-wise RGB agreement against intended digital codes increased from 85.02% in V13 to 94.68% in V14, a 9.66 percentage-point improvement. These values quantify agreement with nominal digital references under the experimental conditions, not certified colourimetric accuracy, because the printed swatches were not independently characterized and some reference colours informed calibration development. The results demonstrate a physical–digital workflow requiring independent validation.

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Last Modified
2026-11-01 20:33:46 GMT+5:30

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Submission Files
CHROMA Colour Intelligence_Bridging Physical Colours and Digital Design.pdf (734.6 Kb, 2026-11-01 20:31:36 GMT+5:30)